

Chapter 10 Study Sheet

Circles — everything you need on one page.

10.1 Lines and Segments

- Chord: endpoints on the circle · diameter: chord through the center · secant: LINE hitting twice · tangent: touches once.
- Tangent $\perp$ radius at the point of tangency; two tangent segments from one point are congruent.
- Find-r setup: $\text{tangent}^2 + r^2 = (r + \text{external})^2$ — expand, cancel r^2 , solve.

10.2 Arc Measures

- Minor arc = central angle · major arc = $360^\circ - \text{minor}$ · semicircle = 180° (three-letter name).
- Arc Addition: adjacent arcs add. Circle graphs: percent $\times 360^\circ$.
- Congruent arcs need congruent circles; ALL circles (and same-measure arcs) are similar.

10.3 Chords

- Congruent chords $\leftrightarrow$ congruent arcs $\leftrightarrow$ equidistant from center.
- Diameter $\perp$ chord bisects the chord and its arc.
- $r^2 = (\text{half-chord})^2 + (\text{distance})^2$ — draw the radius to the endpoint, every time.

10.4 Inscribed Angles

- Inscribed angle = $\frac{1}{2}$ its arc; same arc $\rightarrow$ same angle.
- Semicircle $\rightarrow 90^\circ$ angle. Inscribed quadrilateral $\rightarrow$ opposite angles supplementary.

10.5 Angle Locations

- Vertex on the circle: $\frac{1}{2}$ arc · inside: $\frac{1}{2}(\text{sum})$ · outside: $\frac{1}{2}(\text{far} - \text{near})$ · circumscribed: $180^\circ - \text{central}$.

10.6 Segment Products

- Chords: $(\text{piece})(\text{piece}) = (\text{piece})(\text{piece})$ · secants: $(\text{whole})(\text{external}) = (\text{whole})(\text{external})$ · tangent $^2 = (\text{whole})(\text{external})$.

10.7 Coordinate Plane

- $(x - h)^2 + (y - k)^2 = r^2$. Radius by distance formula; center of a diameter by midpoint; point tests by plugging in.